SHARNA HOSSAIN

Software Engineer

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PROFESSIONAL SUMMARY

Software engineer experienced in building and delivering real-time financial systems and distributed architectures. Skilled in designing reliable data and payment flows, integrating across systems, and translating complex business requirements into production-ready solutions, with a focus on scalability, correctness, and performance.

WORK EXPERIENCE

Software Engineer II

Morgan Stanley | August 2024 - Present | New York, NY

- Led end-to-end production delivery of a real-time payments system, ensuring reliable settlement processing and supporting high-value financial transactions tied to a broader initiative projected to avoid $70M in future costs.
- Designed and implemented distributed system architecture for financial workflows, using ADRs and C4 diagrams to define service boundaries and support scalable, maintainable integrations.
- Developed and supported real-time financial workflows using event-driven patterns, enabling reliable communication between services, improving system resilience, and collaborating closely with stakeholders to align on requirements and system behavior.
- Supported a high-priority initiative enabling real-time payments under a new trade model, ensuring accurate and reliable execution of daily FX transactions in the millions.
- Renovated a legacy end-of-life GWT application with tightly coupled front-end and back-end components into a distributed web platform, modernizing the user interface and service layer to support long-term scalability and maintainability for a high-stakes trading platform handling millions in daily transactions.

Software Engineer I

Morgan Stanley | February 2024 - August 2024 | New York, NY

- Developed a microservice to support real-time settlement processing and financial transaction workflows, enabling reliable communication between internal systems and external banking partners.
- Implemented payment message flows and journal creation logic to ensure accurate tracking and execution of financial transactions.
- Produced architectural decision records and system design documentation, including C4 diagrams, to support scalable system design and cross-team alignment.
- Resolved data integrity issues in upstream systems, improving accuracy and reliability of financial data used across services.

EDUCATION

Bachelor of Arts, Computer Science | Columbia University | 2022

Associate in Science, Computer Science | Borough of Manhattan Community College | 2021

Coding Bootcamp Certificate | New York Code + Design Academy | 2019

SKILLS & TECHNOLOGIES

Languages: Java, SQL, Python, JavaScript, R

Frameworks & Systems: Spring Boot, Angular, RESTful APIs, Distributed Systems

Tools & Platforms: Apache Kafka, IBM MQ, Docker, Jenkins, Snowflake, Gradle, Linux, Git